Unit 4 Question Bank dstl

Q1. Inclusion-Exclusion Principle

In a class C of 2092 students, 1232 are pursuing a course in Spanish, 879 in French, and 114 in Russian. Further, 103 are pursuing courses in both Spanish and French, 23 in both Spanish and Russian, and 14 in both French and Russian. Assuming that each student is pursuing at least one of the three language courses, find how many students are pursuing a course in all three languages.

Solution:

Let

$$S = \{\text{students studying Spanish}\},$$

$$F = \{\text{students studying French}\},$$

$$R = \{\text{students studying Russian}\}.$$

Given:

$$|S| = 1232, \quad |F| = 879, \quad |R| = 114,$$

$$|S \cap F| = 103, \quad |S \cap R| = 23, \quad |F \cap R| = 14.$$

Since every student is studying at least one of the three languages,

$$|S \cup F \cup R| = 2092.$$

By the Inclusion-Exclusion Principle,

$$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |F \cap R| + |S \cap F \cap R|.$$

Therefore,

$$2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|.$$

Hence,

$$2092 = 2085 + |S \cap F \cap R|,$$

so

$$\boxed{|S \cap F \cap R| = 7}.$$

Thus, $\boxed{7}$ students are studying all three languages.

Q2. Integers Divisible by 7 and 11

How many integers between 1000 and 9999 are divisible by 7 and 11?

Solution:

An integer divisible by both 7 and 11 must be divisible by

$$\text{lcm}(7, 11) = 77.$$

The smallest multiple of 77 greater than or equal to 1000 is

$$77 \left\lceil \frac{1000}{77} \right\rceil = 77(13) = 1001.$$

The largest multiple of 77 less than or equal to 9999 is

$$77 \left\lfloor \frac{9999}{77} \right\rfloor = 77(129) = 9933.$$

Therefore, the number of multiples is

$$129 - 13 + 1 = 117.$$

Hence,

$$\boxed{117}$$

integers satisfy the condition.

Q3. Integers Not Divisible by 2, 3, 5 or 7

How many integers between 1 and 500 are not divisible by 2, 3, 5 or 7?

Solution:

Let A, B, C, D denote the sets of integers divisible by $2, 3, 5, 7$, respectively.

There are

$$500$$

integers in total.

Using the Inclusion-Exclusion Principle,

$$|A| = \left\lfloor \frac{500}{2} \right\rfloor = 250,$$

$$|B| = \left\lfloor \frac{500}{3} \right\rfloor = 166,$$

$$|C| = 100, \qquad |D| = 71.$$

The pairwise intersections are

$$|A \cap B| = \left\lfloor \frac{500}{6} \right\rfloor = 83,$$

$$|A \cap C| = \left\lfloor \frac{500}{10} \right\rfloor = 50,$$

$$|A \cap D| = \left\lfloor \frac{500}{14} \right\rfloor = 35,$$

$$|B \cap C| = \left\lfloor \frac{500}{15} \right\rfloor = 33,$$

$$|B \cap D| = \left\lfloor \frac{500}{21} \right\rfloor = 23,$$

$$|C \cap D| = \left\lfloor \frac{500}{35} \right\rfloor = 14.$$

The triple intersections are

$$|A \cap B \cap C| = \left\lfloor \frac{500}{30} \right\rfloor = 16,$$

$$|A \cap B \cap D| = \left\lfloor \frac{500}{42} \right\rfloor = 11,$$

$$|A \cap C \cap D| = \left\lfloor \frac{500}{70} \right\rfloor = 7,$$

$$|B \cap C \cap D| = \left\lfloor \frac{500}{105} \right\rfloor = 4.$$

Finally,

$$|A \cap B \cap C \cap D| = \left\lfloor \frac{500}{210} \right\rfloor = 2.$$

Thus the number divisible by at least one of $2, 3, 5, 7$ is

$$\begin{aligned} &250 + 166 + 100 + 71 \\ &- (83 + 50 + 35 + 33 + 23 + 14) \\ &+ (16 + 11 + 7 + 4) - 2 \\ &= 386. \end{aligned}$$

Therefore, the number not divisible by any of $2, 3, 5, 7$ is

$$500 - 386 = 114.$$

Hence,

$$\boxed{114}.$$

Q4. Pigeonhole Principle

Use the pigeonhole principle to justify that, among 11 numbers selected randomly between 1 and 330, there are at least two numbers such that their difference is less than or equal to 38.

Solution:

Divide the numbers from 1 to 330 into 10 groups as follows:

$$P_1 = \{1, 2, \ldots, 33\}, \qquad P_2 = \{34, 35, \ldots, 66\},$$
$$P_3 = \{67, \ldots, 99\}, \qquad \cdots$$
$$P_{10} = \{298, \ldots, 330\}.$$

Thus, there are 10 pigeonholes and 11 selected numbers.

By the Pigeonhole Principle, at least two of the 11 numbers must lie in the same group.

Any two numbers in the same group differ by at most

$$33 - 1 = 32.$$

Therefore their difference is certainly less than or equal to 38.

Hence, among any 11 numbers selected from 1 to 330, there are at least two numbers whose difference is at most 38.

Proved.

Q5. Pigeonhole Principle and Friends

Use the pigeonhole principle to justify that, among six people at a party, at least three are mutually friends or at least three are mutually strangers.

Solution:

Choose any one person, say P.

There are five other people. Each of these five people is either a friend of P or a stranger to P.

By the Pigeonhole Principle, at least three of these five people must belong to the same category.

Case 1: At least three are friends of P

Suppose A, B, C are all friends of P.

If any two among A, B, C are friends, say A and B, then

$$P, A, B$$

are three mutually friends.

If no two among A, B, C are friends, then

$$A, B, C$$

are three mutually strangers.

Case 2: At least three are strangers to P

The same argument applies. If any two of these three people are strangers, together with P they form three mutually strangers. Otherwise, the three people themselves are mutually friends.

Therefore, among any six people, there are either three mutual friends or three mutual strangers.

Proved.

Q6. Abelian Monoid on $\mathbb{Q}$

On the set of rational numbers $\mathbb{Q}$, define

$$a * b = a + b + ab, \qquad a, b \in \mathbb{Q}.$$

Show that $(\mathbb{Q}, *)$ is an Abelian monoid and find all invertible elements.
Solution:

Closure

For $a, b \in \mathbb{Q}$,

$$a + b + ab \in \mathbb{Q}.$$

Therefore, $*$ is closed on $\mathbb{Q}$.

Associativity

We have

$$\begin{aligned}(a * b) * c &= (a + b + ab) + c + (a + b + ab)c \\ &= a + b + c + ab + ac + bc + abc.\end{aligned}$$

Also,

$$\begin{aligned}a * (b * c) &= a + (b + c + bc) + a(b + c + bc) \\ &= a + b + c + ab + ac + bc + abc.\end{aligned}$$

Hence,

$$(a * b) * c = a * (b * c).$$

Thus, $*$ is associative.

Commutativity

$$a * b = a + b + ab = b + a + ba = b * a.$$

Thus, $*$ is commutative.

Identity element

Let e be the identity. Then

$$a * e = a.$$

Thus,

$$a + e + ae = a,$$

which gives

$$e(1 + a) = 0.$$

For all a, we obtain

$$\boxed{e = 0}.$$

Indeed,

$$a * 0 = a.$$

Therefore, $(\mathbb{Q}, *)$ is a monoid, and since it is also commutative, it is an Abelian monoid.

Invertible elements

An element a is invertible if there exists $b \in \mathbb{Q}$ such that

$$a * b = 0.$$

Therefore,

$$a + b + ab = 0.$$

Hence,

$$b(1 + a) = -a.$$

If $a \neq -1$,

$$\boxed{b = -\frac{a}{1+a}}.$$

If $a = -1$, then

$$-1 + b - b = -1 \neq 0.$$

Thus -1 has no inverse.
Hence all invertible elements are

$$\boxed{\mathbb{Q} \setminus \{-1\}},$$

with

$$\boxed{a^{-1_*} = -\frac{a}{1+a}}.$$

Q7. Non-Commutative Monoid

On

$$M = \mathbb{Q} \times \mathbb{Q},$$

define

$$(a, b) * (x, y) = (ax, ay + b).$$

Show that $(M, *)$ is a non-commutative monoid and find all invertible elements.
Solution:

Closure

For $(a, b), (x, y) \in M$,

$$(ax, ay + b) \in \mathbb{Q} \times \mathbb{Q}.$$

Hence $*$ is closed.

Associativity

Let $(a,b),(x,y),(u,v) \in M$.

Then

$$\begin{aligned}((a,b)*(x,y))*(u,v) &= (ax, ay+b)*(u,v)\\ &= (axu, axv+ay+b).\end{aligned}$$

On the other hand,

$$\begin{aligned}(a,b)*((x,y)*(u,v)) &= (a,b)*(xu, xv+y)\\ &= (axu, a(xv+y)+b)\\ &= (axu, axv+ay+b).\end{aligned}$$

Therefore,

$$((a,b)*(x,y))*(u,v) = (a,b)*((x,y)*(u,v)).$$

Thus $*$ is associative.

Identity element

Let (e_1,e_2) be the identity. We require

$$(a,b)*(e_1,e_2) = (a,b).$$

Thus,

$$(ae_1, ae_2+b) = (a,b).$$

Hence

$$e_1 = 1, \qquad e_2 = 0.$$

Therefore, the identity is

$$\boxed{(1,0)}.$$

Non-commutativity

For example,

$$(2,1)*(3,4) = (6,9),$$

whereas

$$(3,4)*(2,1) = (6,7).$$

Since

$$(6,9) \neq (6,7),$$

the operation is not commutative.

Invertible elements

Let (a, b) have inverse (x, y). Then

$$(a, b) * (x, y) = (1, 0).$$

Thus,

$$ax = 1,$$

and

$$ay + b = 0.$$

Therefore,

$$x = \frac{1}{a},$$

which requires

$$a \neq 0.$$

Also,

$$y = -\frac{b}{a}.$$

Hence,

$$\boxed{(a, b)^{-1} = \left(\frac{1}{a}, -\frac{b}{a}\right)}$$

for all

$$\boxed{a \neq 0}.$$

Thus the invertible elements are

$$\boxed{\{(a, b) \in \mathbb{Q} \times \mathbb{Q} : a \neq 0\}}.$$

Q8. Abelian Group on $\mathbb{Q}^+$

On $\mathbb{Q}^+$, define

$$a * b = \frac{ab}{3}.$$

Show that $(\mathbb{Q}^+, *)$ is an Abelian group.
Solution:

Closure

For $a, b \in \mathbb{Q}^+$,

$$\frac{ab}{3} \in \mathbb{Q}^+.$$

Thus the operation is closed.

Associativity

$$\begin{aligned}(a * b) * c &= \frac{ab}{3} * c \\ &= \frac{abc}{9}.\end{aligned}$$

Similarly,

$$\begin{aligned}a * (b * c) &= a * \frac{bc}{3} \\ &= \frac{abc}{9}.\end{aligned}$$

Therefore,

$$(a * b) * c = a * (b * c).$$

Identity

Let e be the identity. Then

$$a * e = a.$$

Therefore,

$$\frac{ae}{3} = a,$$

so

$$e = 3.$$

Thus the identity element is

$$\boxed{3}.$$

Inverse

For $a \in \mathbb{Q}^+$, let b be its inverse. Then

$$a * b = 3.$$

Hence

$$\frac{ab}{3} = 3,$$

so

$$b = \frac{9}{a}.$$

Therefore,

$$\boxed{a^{-1} = \frac{9}{a}}.$$

Commutativity

$$a * b = \frac{ab}{3} = \frac{ba}{3} = b * a.$$

Therefore $(\mathbb{Q}^+, *)$ is an Abelian group.

$$\boxed{(\mathbb{Q}^+, *) \text{ is an Abelian group.}}$$

Q9. Multiplication Tables of $\mathbb{Z}_4$

Write the multiplication tables for $\mathbb{Z}_4$ under addition modulo 4 and multiplication modulo 4. Determine whether $(\mathbb{Z}_4, +_4)$ is a group and show that $(\mathbb{Z}_4 \setminus \{0\}, \times_4)$ is not a group.

Solution:

We have

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}.$$

Addition modulo 4

$+_4$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

The identity is 0.

The inverses are

$$0^{-1} = 0, \qquad 1^{-1} = 3, \qquad 2^{-1} = 2, \qquad 3^{-1} = 1.$$

Therefore,

$$\boxed{(\mathbb{Z}_4, +_4) \text{ is a group.}}$$

It is also Abelian because addition modulo 4 is commutative.

Multiplication modulo 4

$\times_4$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

Consider

$$\mathbb{Z}_4 \setminus \{0\} = \{1, 2, 3\}.$$

Although 1 and 3 have multiplicative inverses,

$$1^{-1} = 1, \qquad 3^{-1} = 3,$$

the element 2 has no multiplicative inverse because

$$2 \times_4 1 = 2,$$

$$2 \times_4 2 = 0,$$

$$2 \times_4 3 = 2.$$

Thus there is no $x \in \{1, 2, 3\}$ such that

$$2x \equiv 1 \pmod 4.$$

Hence,

$$\boxed{(\mathbb{Z}_4 \setminus \{0\}, \times_4) \text{ is not a group.}}$$

Q10. Cyclic Group C_4

Let $i = \sqrt{-1}$. Show that

$$C_4 = \{1, -1, i, -i\}$$

is a cyclic group under usual multiplication. Find all generators.

Solution:

We have

$$i^2 = -1, \qquad i^3 = -i, \qquad i^4 = 1.$$

Therefore,

$$\langle i \rangle = \{1, i, i^2, i^3\} = \{1, i, -1, -i\}.$$

Thus,

$$C_4 = \langle i \rangle.$$

Hence C_4 is cyclic.

Since

$$(-i)^2 = -1, \qquad (-i)^3 = i, \qquad (-i)^4 = 1,$$

we also have

$$\langle -i \rangle = C_4.$$

The elements 1 and -1 have orders 1 and 2, respectively, while i and $-i$ have order 4.

Therefore, the generators are

$$\boxed{i \text{ and } -i}.$$

Q11. Properties of Inverses

Let $(G, *)$ be a group. Prove:

1. $(a^{-1})^{-1} = a$;
2. $(a * b)^{-1} = b^{-1} * a^{-1}$.

Proof:

1. Proof that $(a^{-1})^{-1} = a$

Since a^{-1} is the inverse of a,

$$a * a^{-1} = a^{-1} * a = e.$$

Therefore, a is the inverse of a^{-1}.

Hence,

$$\boxed{(a^{-1})^{-1} = a}.$$

2. Proof that $(a * b)^{-1} = b^{-1} * a^{-1}$

Consider

$$(a * b) * (b^{-1} * a^{-1}).$$

By associativity,

$$\begin{aligned}(a * b) * (b^{-1} * a^{-1}) &= a * (b * b^{-1}) * a^{-1} \\ &= a * e * a^{-1} \\ &= a * a^{-1} \\ &= e.\end{aligned}$$

Similarly,

$$\begin{aligned}(b^{-1} * a^{-1}) * (a * b) &= b^{-1} * (a^{-1} * a) * b \\ &= b^{-1} * e * b \\ &= b^{-1} * b \\ &= e.\end{aligned}$$

Thus $b^{-1} * a^{-1}$ is the inverse of $a * b$.

Therefore,

$$\boxed{(a * b)^{-1} = b^{-1} * a^{-1}}.$$

Q12. Intersection and Union of Subgroups

Show that the intersection of any two subgroups of a group is a subgroup. Give an example showing that the union of two subgroups need not be a subgroup.

Solution:

Let H and K be subgroups of G.

We prove that

$$H \cap K$$

is a subgroup of G.

Since H and K are subgroups,

$$e \in H, \qquad e \in K.$$

Hence,

$$e \in H \cap K.$$

Let $a, b \in H \cap K$. Then

$$a, b \in H$$

and

$$a, b \in K.$$

Since H and K are subgroups,

$$ab^{-1} \in H$$

and

$$ab^{-1} \in K.$$

Therefore,

$$ab^{-1} \in H \cap K.$$

By the subgroup test,

$$\boxed{H \cap K \leq G}.$$

Union need not be a subgroup

Consider

$$G = (\mathbb{Z}, +).$$

Let

$$H = 2\mathbb{Z}$$

and

$$K = 3\mathbb{Z}.$$

Both H and K are subgroups of $\mathbb{Z}$.

However,

$$2 \in H \subseteq H \cup K$$

and

$$3 \in K \subseteq H \cup K.$$

But

$$2 + 3 = 5 \notin H \cup K,$$

since 5 is neither divisible by 2 nor by 3.

Therefore,

$$H \cup K$$

is not a subgroup.

Hence,

The union of two subgroups need not be a subgroup.

Q13. Cosets of $5\mathbb{Z}$

Compute all cosets of

$$H = 5\mathbb{Z}$$

in $(\mathbb{Z}, +)$ and find $[G : H]$.

Solution:

We have

$$H = 5\mathbb{Z} = \{\ldots, -10, -5, 0, 5, 10, \ldots\}.$$

The distinct cosets are

$$0 + H = 5\mathbb{Z},$$

$$1 + H = \{\ldots, -9, -4, 1, 6, 11, \ldots\},$$

$$2 + H = \{\ldots, -8, -3, 2, 7, 12, \ldots\},$$

$$3 + H = \{\ldots, -7, -2, 3, 8, 13, \ldots\},$$

$$4 + H = \{\ldots, -6, -1, 4, 9, 14, \ldots\}.$$

Thus the complete set of distinct cosets is

$$\boxed{5\mathbb{Z},\ 1+5\mathbb{Z},\ 2+5\mathbb{Z},\ 3+5\mathbb{Z},\ 4+5\mathbb{Z}}.$$

Therefore,

$$\boxed{[\mathbb{Z}:5\mathbb{Z}]=5}.$$

Q14. Conjugation in S_6

Let

$$\alpha=(13562), \qquad \beta=(1523)(46)\in S_6.$$

Compute

$$\alpha^{-1}\circ\beta\circ\alpha$$

and

$$\beta^{-1}\circ\alpha\circ\beta.$$

Solution:
We use the conjugation rule

$$\sigma^{-1}(a_1\,a_2\,\ldots\,a_k)\sigma=(\sigma^{-1}(a_1)\,\sigma^{-1}(a_2)\,\ldots\,\sigma^{-1}(a_k)).$$

For the first expression,

$$\alpha^{-1}\beta\alpha=\alpha^{-1}(1523)(46)\alpha.$$

Since

$$\alpha^{-1}(1)=2,\quad \alpha^{-1}(5)=4,\quad \alpha^{-1}(2)=3,\quad \alpha^{-1}(3)=6,$$

and

$$\alpha^{-1}(4)=5,\quad \alpha^{-1}(6)=1,$$

we obtain

$$\boxed{\alpha^{-1}\beta\alpha=(2436)(51)}.$$

Since cycles may be written starting from any element, this can also be written as

$$\boxed{(1236)(45)}.$$

For the second expression,

$$\beta^{-1}\alpha\beta.$$

We apply β^{-1} to the entries of $\alpha=(13562)$.
Since

$$\beta^{-1}(1)=2,\quad \beta^{-1}(3)=2?$$

More directly, using the permutation mappings,

$$\beta=(1523)(46),$$

we have

$$\beta^{-1}(1)=3,\quad \beta^{-1}(3)=2,\quad \beta^{-1}(5)=1,\quad \beta^{-1}(6)=4,\quad \beta^{-1}(2)=5.$$

Thus,

$$\boxed{\beta^{-1}\alpha\beta = (32154)}.$$

Equivalently, reversing the starting point,

$$\boxed{(15432)}.$$

Therefore,

$$\boxed{\alpha^{-1}\beta\alpha = (1236)(45)}$$

and

$$\boxed{\beta^{-1}\alpha\beta = (15432)}.$$

Q15. Lagrange's Theorem

Theorem: If G is a finite group and H is a subgroup of G, then

$$|H| \mid |G|.$$

Moreover,

$$|G| = [G : H]|H|.$$

Proof:
Let the distinct left cosets of H in G be

$$a_1H, a_2H, \ldots, a_kH.$$

Every coset a_iH has the same number of elements as H because the map

$$H \to a_iH, \qquad h \mapsto a_ih$$

is a bijection.
Therefore,

$$|a_iH| = |H|.$$

The left cosets are pairwise disjoint and together contain all elements of G. Hence,

$$|G| = |a_1H| + \cdots + |a_kH| = k|H|.$$

Since $k = [G : H]$,

$$\boxed{|G| = [G : H]|H|}.$$

Therefore,

$$\boxed{|H| \mid |G|}.$$

Deduction: Every group of prime order is cyclic

Let $|G| = p$, where p is prime.

Choose any

$$a \in G, \qquad a \neq e.$$

The subgroup $\langle a \rangle$ is a subgroup of G. By Lagrange's Theorem,

$$|\langle a \rangle| \mid p.$$

Since p is prime,

$$|\langle a \rangle| = 1 \quad \text{or} \quad |\langle a \rangle| = p.$$

But $a \neq e$, so

$$|\langle a \rangle| \neq 1.$$

Hence,

$$|\langle a \rangle| = p.$$

Thus,

$$\langle a \rangle = G.$$

Therefore,

$$\boxed{\text{Every group of prime order is cyclic.}}$$

Q16. Characterization of Normal Subgroups

Let H be a subgroup of a group G. Show that H is normal if and only if

$$g^{-1}hg \in H, \qquad \forall h \in H, \quad g \in G.$$

Proof:

Suppose first that H is normal in G.

By definition,

$$gH = Hg$$

for every $g \in G$.

Therefore, for every $h \in H$,

$$g^{-1}hg \in H.$$

Thus,

$$\boxed{g^{-1}hg \in H}.$$

Conversely, suppose that

$$g^{-1}hg \in H$$

for every $h \in H$ and $g \in G$.

Then

$$g^{-1}Hg \subseteq H.$$

Replacing g by g^{-1} gives

$$gHg^{-1} \subseteq H.$$

Multiplying by g^{-1} on the left and g on the right gives

$$H \subseteq g^{-1}Hg.$$

Therefore,

$$g^{-1}Hg = H.$$

Hence,

$$gH = Hg$$

for all $g \in G$.

Thus,

$$\boxed{H \trianglelefteq G}.$$

Therefore,

$$\boxed{H \trianglelefteq G \iff g^{-1}hg \in H \quad \forall h \in H,\ g \in G.}$$

Q17. Normal Subgroups of S_3

Let

$$H = \{e, (23)\} \leq S_3,$$

where

$$S_3 = \{e, (12), (13), (23), (123), (132)\}.$$

Show that H is not normal in S_3. Further, show that

$$K = \{e, (123), (132)\}$$

is normal in S_3.

Solution:

Showing that H is not normal

Take

$$g = (12) \in S_3$$

and

$$h = (23) \in H.$$

Then

$$g^{-1}hg = (12)(23)(12).$$

Conjugation changes the symbols according to (12):

$$(12)(23)(12) = (13).$$

But

$$(13) \notin H.$$

Therefore,

$$g^{-1}hg \notin H.$$

Hence,

$$\boxed{H \text{ is not normal in } S_3}.$$

Showing that K is normal

We have

$$K = \{e, (123), (132)\}.$$

The subgroup K has order 3, while

$$|S_3| = 6.$$

The index is

$$[S_3 : K] = \frac{6}{3} = 2.$$

Every subgroup of index 2 is normal. Therefore,

$$\boxed{K \trianglelefteq S_3}.$$

Alternatively, K consists of the identity and all 3-cycles in S_3, and conjugation preserves cycle type. Hence every conjugate of (123) or (132) remains in K.

Thus,

$$\boxed{K \text{ is normal in } S_3}.$$

Q18. Definitions in Ring Theory

Define the following and give an example of each.

1. Ring with unity 1

A ring R is said to have a unity (or multiplicative identity) 1 if there exists an element $1 \in R$ such that

$$1a = a1 = a$$

for every $a \in R$.

Example:

$$(\mathbb{Z}, +, \times)$$

is a ring with unity 1.

2. Non-commutative ring

A ring R is called non-commutative if there exist $a, b \in R$ such that

$$ab \neq ba.$$

Example: The ring of 2×2 matrices

$$M_2(\mathbb{R})$$

is non-commutative.

For example, let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

whereas

$$BA = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus

$$AB \neq BA.$$

3. Zero-divisor in a commutative ring

Let R be a commutative ring. A nonzero element $a \in R$ is called a zero-divisor if there exists a nonzero element $b \in R$ such that

$$ab = 0.$$

Example:
In

$$\mathbb{Z}_6,$$

we have

$$2 \neq 0, \qquad 3 \neq 0,$$

but

$$2 \cdot 3 = 6 \equiv 0 \pmod{6}.$$

Hence 2 and 3 are zero-divisors in $\mathbb{Z}_6$.

4. Unit of a ring with unity 1

Let R be a ring with unity 1. An element $a \in R$ is called a unit if there exists $b \in R$ such that

$$ab = ba = 1.$$

The element b is called the multiplicative inverse of a.
Example:
In $\mathbb{Z}$, the units are

$$1 \quad \text{and} \quad -1,$$

because

$$1(1) = 1$$

and

$$(-1)(-1) = 1.$$

5. Field

A field is a commutative ring with unity $1 \neq 0$ in which every nonzero element has a multiplicative inverse.

Example:

The rational numbers

$$\mathbb{Q}$$

form a field under ordinary addition and multiplication.

For every nonzero

$$\frac{a}{b} \in \mathbb{Q},$$

its multiplicative inverse is

$$\frac{b}{a}.$$

Thus,

$$\boxed{\mathbb{Q} \text{ is a field.}}$$

Q19. Finite Integral Domain

Prove that every finite integral domain is a field.

Proof:

Let D be a finite integral domain.

We need to show that every nonzero element of D has a multiplicative inverse.

Let

$$a \in D, \qquad a \neq 0.$$

Consider the function

$$f : D \to D$$

defined by

$$f(x) = ax.$$

We first show that f is injective.

Suppose

$$f(x) = f(y).$$

Then

$$ax = ay.$$

Therefore,

$$a(x - y) = 0.$$

Since D is an integral domain, it has no zero-divisors. Since $a \neq 0$, we must have

$$x - y = 0.$$

Hence,

$$x = y.$$

Therefore, f is injective.

Since D is finite, every injective function from D to itself is also surjective.

Thus there exists some $x \in D$ such that

$$f(x) = 1.$$

Hence,

$$ax = 1.$$

Since D is commutative,

$$xa = ax = 1.$$

Therefore, x is the multiplicative inverse of a.
Thus every nonzero element of D has a multiplicative inverse.
Therefore,

$$\boxed{\text{Every finite integral domain is a field.}}$$